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Research Article

Lattice Oxygen Refilling for Stable Acidic Water Oxidation

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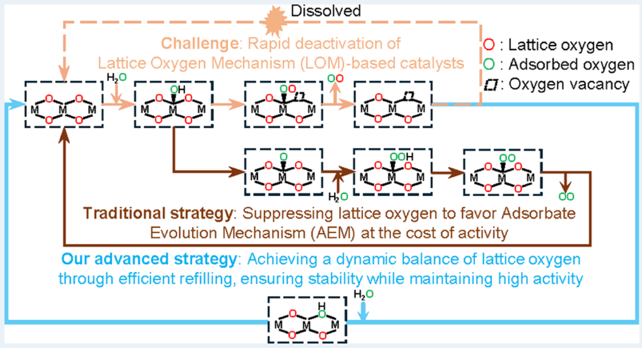
ABSTRACT: Lattice oxygen mechanism (LOM)-based catalysts, exemplified by RuO_2, exhibit a high intrinsic activity for the oxygen evolution reaction (OER). However, their practical application is severely limited by poor stability due to structural degradation during operation. Conventional stabilization strategies, which reduce Ru−O covalency to enhance stability, often suppress lattice oxygen reactivity, shifting the reaction pathway from the efficient LOM to the less effective adsorbate evolution mechanism (AEM), thereby sacrificing activity. This study introduces a (Ce W)-RuO_2 catalyst designed to leverage a lattice oxygen refilling mechanism (LORM). This approach effectively balances lattice oxygen consumption and regeneration, addressing the long-standing trade-off between activity and stability. Using Raman spectroscopy and a simulated slow-growth process, we confirm the operational stability and mechanism of the catalyst. The (Ce W)-RuO_2 catalyst achieves improved stability under acidic conditions without compromising its intrinsic activity, offering a breakthrough solution for the OER in energy conversion technologies.



KEYWORDS: *water oxidation, RuO_2-based catalyst, performance trade-off, lattice oxygen, redox properties*

1. INTRODUCTION

Hydrogen plays a critical role in advancing sustainability by providing a clean, flexible, and scalable energy solution that can support the transition to a low-carbon economy and address various environmental and energy challenges.[1−3] Among the technologies for hydrogen production via water splitting, proton-exchange membrane water electrolyzers (PEMWEs) stand out for their compact design, high current density, and ability to handle intermittent renewable energy sources like wind and solar.[4−7] However, robust anodic catalysts are essential to ensure stable operation in the highly corrosive PEMWE environment.[8] RuO_2 is a promising catalyst for the oxygen evolution reaction (OER) due to its affordability and excellent activity. However, its stability is hindered by rapid dissolution under acidic conditions.[9,10] Specifically, RuO_2 catalyzes the OER via the lattice oxygen mechanism (LOM)[11,12] (Figure 1a), which differs from the conventional adsorbate evolution mechanism (AEM)[13,14] (Figure 1b). This LOM pathway enables RuO_2 to achieve faster OER kinetics by forming O_2 directly from lattice oxygen, bypassing the *OOH intermediate and overcoming the linear limitation of adsorption energy.[15−18] However, rapid lattice oxygen consumption and sluggish lattice oxygen refilling lead to the formation of the soluble high-valence intermediate $*V_O$-RuO_4^{2-} (V_O denotes oxygen vacancy).[19−21] Various strategies have attempted to mitigate this issue, including heteroatom doping,[22−24] heterostructure design,[25−27] and defect engineering.[28,29] These approaches shift the reaction pathway from the LOM to the AEM by weakening Ru−O interactions and reducing Ru−O covalency.[11] While the shift of the reaction pathway improves RuO_2 stability, it compromises intrinsic activity and fails to address the core stability issue of the LOM. Here, we introduce a new strategy to enhance LOM stability by balancing lattice oxygen consumption and refilling (Figure 1c) rather than replacing the LOM pathway with the AEM. By qualitatively evaluating the lattice oxygen refilling rate, we optimize materials to promote lattice oxygen refilling, overcoming the long-standing trade-off between stability and activity in RuO_2-based catalysts.

The unique redox properties of the nonstoichiometric ceria (CeO_x), driven by the facile Ce^{3+}/Ce^{4+} transition, endow it with exceptional oxygen buffering capacity,[30,31] offering significant potential to mitigate lattice oxygen degradation. While CeO_x has been widely employed to optimize electronic structures[32,33] or regulate lattice oxygen activity[34,35] in various catalytic reactions, its application to improve LOM-OER

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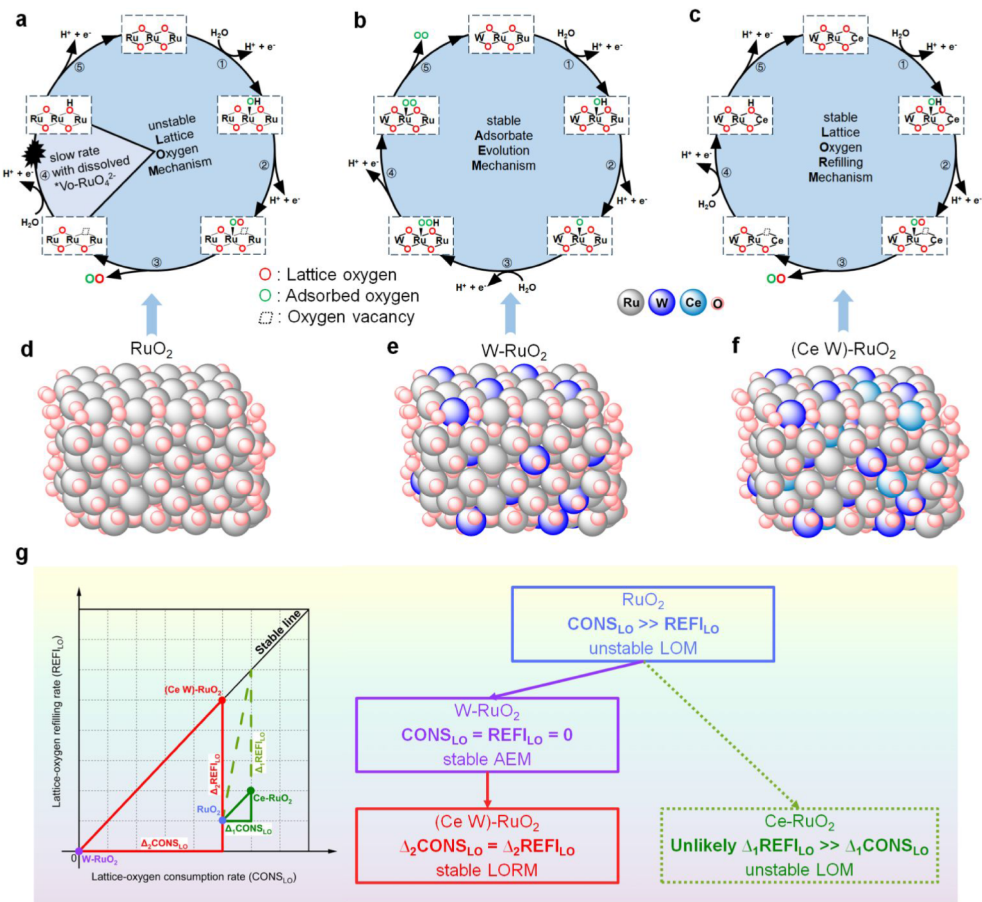

Figure 1. The OER pathways and design principle for catalysts. (a) The unstable LOM pathway in the RuO_2 catalyst. (b) The AEM pathway in the W-RuO_2 catalyst with improved stability but low intrinsic activity. (c) The LORM pathway in the (Ce W)-RuO_2 catalyst with improved stability and high intrinsic activity. Schematic atomic structures of (d) RuO_2, (e) W-RuO_2, and (f) (Ce W)-RuO_2. (g) The design principle for catalysts with balanced lattice oxygen consumption and refilling.

stability remains unexplored. The primary challenge lies in the dual role of redox-active CeO_x, as it facilitates both lattice oxygen consumption and refilling, making it difficult to achieve a steady balance. Herein, we address this challenge by employing an AEM-OER catalyst as an intermediary to overcome the poor selectivity of CeO_x for lattice oxygen refilling. This approach successfully balances lattice oxygen consumption and refilling, enabling a stable LOM-OER for RuO_2. Specifically, RuO_2 (atomic structure in Figure 1d and blue dot in Figure 1g) suffers from instability because its lattice oxygen consumption ($CONS_{LO}$) rate significantly exceeds its lattice oxygen refilling ($REFI_{LO}$) rate. When Ce is doped into RuO_2 (green dot in Figure 1g), a substantial increase in Δ_1REFI_{LO} relative to Δ_1CONS_{LO} is required to reach the stability line (diagonal), which is challenging. To address this, we first synthesized the W-doped RuO_2 (W-RuO_2) catalyst (atomic structure in Figure 1e and purple dot in Figure 1g), which stabilizes the OER via the AEM pathway. Subsequently, we developed a Ce- and W-codoped RuO_2 ((Ce W)-RuO_2) catalyst (atomic structure in Figure 1f and red dot in Figure 1g) with an optimal Ce doping amount. This catalyst reactivates lattice oxygen, achieving a balance, as shown by the near-equal increments of Δ_2REFI_{LO} and Δ_2CONS_{LO} (Figure 1g). As a result, the (Ce W)-RuO_2 catalyst demonstrated outstanding OER activity and stability through the lattice oxygen refilling mechanism (LORM) under acidic conditions (Figure 1c), achieving overpotentials of 206 mV at 10 mA/cm^2 and 257 mV at 100 mA/cm^2. During a 550 h chronopotentiometry (CP) test at 10 mA/cm^2, the potential increased at an average rate of only 0.52 mV/day. The superior OER stability of the (Ce W)-RuO_2 catalyst than the RuO_2 catalyst arises from the smaller energy barrier and faster rate of lattice oxygen refilling, which was qualitatively evaluated by slow-growth molecular dynamics simulation and Raman spectroscopy. To our knowledge, this work is the first to quantify the evaluation of lattice oxygen refilling and improve the stability of the OER of RuO_2 by balancing lattice oxygen consumption and refilling. This approach paves the way for designing more robust and efficient catalysts for OER in energy conversion technologies.

2. METHODS

2.1. Catalyst Preparation. Before electrodeposition, the required solutions were prepared in advance. The Ru^{3+} solution was prepared by adding $RuCl_3 \cdot xH_2O$ (80 mg) into diluted HCl solution (0.8 mL of 37% HCl mixed with 3.2 mL of H_2O), and then the solution was stirred thoroughly to prepare $RuCl_3 \cdot xH_2O$ entirely dissolved. The WO_4^{2-} solution was prepared by adding WO_3 (120 mg) and KOH (220 mg) into H_2O (3 mL), and sequentially the solution was heated to 70 °C and stirred vigorously to entirely dissolve WO_3. The K_2SO_4 solution was prepared by dissolving K_2SO_4 (46 mg) into H_2O (5 mL), and the ionic concentration is about 0.05 mol/L. The Ce^{3+} solution was prepared by dissolving

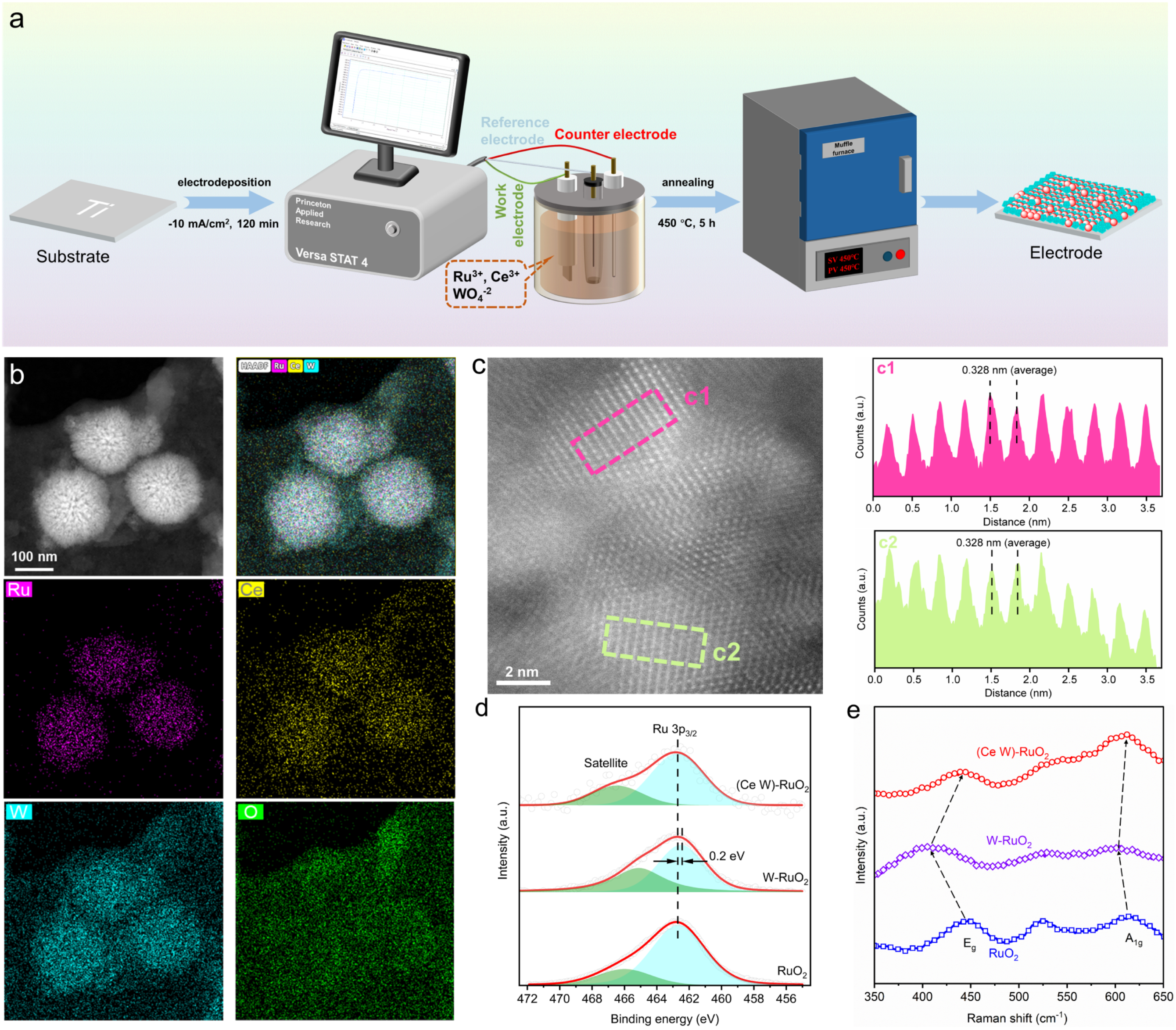

Figure 2. Synthesis and structural characterizations of the (Ce W)-RuO_2 catalyst. (a) Schematic illustration for synthesizing the (Ce W)-RuO_2 catalyst. (b) EDS area scans of RuO_2 spheres by HAADF-STEM. (c) Lattice fringes of the (Ce W)-RuO_2 catalyst by HAADF-STEM. (d) XPS spectra. (e) Raman spectra of RuO_2, W-RuO_2, and (Ce W)-RuO_2.

$Ce(NO_3)_3{\cdot}6H_2O$ (434 mg) into H_2O (5 mL) and the Ce^{3+} concentration is about 0.2 mol/L. To synthesize the (Ce W)-RuO_2 catalyst, the prepared WO_4^{2-} solution (277.5 μL), Ru^{3+} solution (600 μL), K_2SO_4 solution (510 μL), and Ce^{3+} solution (72.3 μL) were added to DI water in turn and stirred thoroughly to obtain the final electrodeposition solution with a total volume of 20 mL. In the three-electrode system, a Ti substrate was used as the working electrode (WE) in the electrodeposition solution and was applied with −10 mA/cm^2 current density for 2 h to form a self-supporting precursor, which afterward was placed in a muffle furnace for annealing (5 h, 450 °C) under an ambient atmosphere. This straightforward method was used to synthesize the (Ce W)-RuO_2 catalyst. All of the comparison samples (RuO_2, W-RuO_2, Ce-RuO_2) were synthesized using the same method, except for the different ionic species added to the electrodeposition solution.

2.2. Catalyst Characterization. Scanning electron microscopy (SEM) and energy-dispersive spectroscopy (EDS) were performed using a JEOL FE-SEM 7800F and FE-SEM SU8010. The high-angle annular dark field scanning transmission electron microscopy (HAADF-STEM) was performed by using a condenser spherical aberration-corrected transmission electron microscope (Spectra 200). The Ru mass loading was determined by using an inductively coupled plasma optical emission spectrometer (ICP-OES, iCAP PRO XP Duo). The Raman spectra were obtained using an inVia confocal Raman microscope equipped with a 50× objective lens and a laser wavelength of 532 nm (the reported spectra are an average of 50 measured spectra).

2.3. Electrochemical Measurements. All electrochemical measurements were performed by using a standard three-electrode system. The self-supporting electrodes were used as the working electrode (WE). The Pt wire was used as the counter electrode (CE), and the calibrated Ag/AgCl electrode was used as the reference electrode (RE). The electrolyte for the OER performance analysis was a sulfuric acid solution with

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a concentration of 0.5 mol/L. The pH value of the electrolyte was tested three times with a pH meter, and the average value of 0.31 was used in this work to calculate the potential vs the reversible hydrogen electrode (RHE). The potentials were calibrated against the RHE according to the following equation: $E_{RHE} = E_{Ag/AgCl} + 0.059\ pH + 0.197$. The scan rate was 10 mV/s for all linear sweep voltammetry and cyclic voltammetry tests. The LSV curves were plotted with *iR* compensation, where the required resistance was determined by electrochemical impedance spectroscopy (EIS). The frequency for EIS measurement ranged from 0.05 Hz to 10^5 Hz at a potential of 1.25 V vs the Ag/AgCl electrode. The Tafel slopes were obtained by fitting the linear stages to the LSV curves. As the scan rate in the LSV tests was 10 mV/s, the fitted Tafel slope should be slightly smaller than the true value for the steady state, and so these Tafel slopes are only compared among our catalysts without reference comparison.

2.4. Computational Method. All spin-polarized computations were performed within the framework of density functional theory (DFT), employing the Vienna ab initio simulation package (VASP). The Perdew−Burke−Ernzerhof (PBE) functional, within a generalized gradient approximation (GGA), was utilized to describe the exchange and correlation effects of electrons. The projector-augmented wave (PAW) method was employed to describe the electron−ion interaction. The kinetic energy cutoff for the plane wave basis was set to 400 eV. The Brillouin zone was sampled by a $2 \times 3 \times 1$ grid for structure relaxation and a $5 \times 8 \times 1$ grid for DOS computation. The criterion for structure relaxation was set to 10^{-5} eV for the total energy and 0.05 eV/Å for the force on each atom. The DFT-D3 method was employed for the correction of vdW forces.

2.5. Catalyst Modeling. A 1×3 supercell of the RuO_2 (110) facet was employed for pristine RuO_2. The bottom two layers were fixed to simulate the bulk phase, and a 20 Å vacuum layer was introduced to avoid the interaction between different layers. The W-RuO_2 and (Ce W)-RuO_2 with doped W and Ce atoms were modeled by replacing a surface Ru atom with a Ce or W atom. The Ru vacancies were constructed by removing the surface Ru atoms.

2.6. Slow-Growth Simulation. The kinetic energy barriers for oxygen vacancy refilling were computed by utilizing ab initio molecular dynamics (AIMD) and the slow-growth method within the canonical (NVT) ensemble, regulated by Nosé−Hoover thermostats at a fixed temperature of 300 K and a time step of 1.0 fs. Twelve explicit H_2O molecules were placed above the O_v-(Ce W)-RuO_2 (or O_v-RuO_2) surface to simulate the effect of the solvent environment. The structures were optimized until the force was less than 0.05 eV/Å, with a cutoff energy of 300 eV set for the kinetic computation procedure. Then, the system was heated to 300 K by recalibrating the velocity every 20 steps, followed by a free AIMD pre-equilibration for 5 ps. The collective variables (CVs) were set to the distance between the O and the (Ru Ru) or (Ru Ce). The slow-growth simulation was then performed with a reaction coordinate change of 0.0005 Å per time step, along with the blue-moon sampling. To obtain the kinetic barriers of elementary steps, the slow-growth method combined with a blue-moon ensemble was introduced. In the blue moon ensemble, we set a constrained reaction coordinate ξ (denoting the distance of Ce−O and Ru−O in this work). Then, the mean force is the average constraint force (represented by the Lagrange multiplier) gathered during the MD simulation under the constraint ξ. Then, the free energy $G(\xi)$ can be calculated by integrating the mean force f_ξ. Briefly, the mean force is the direct output of the slow-growth simulation, validating the reliability of the data, and free energy is the integral of the mean force, illustrating the kinetic barrier.

3. RESULTS AND DISCUSSION

3.1. Material Characterization. The (Ce W)-RuO_2 catalyst was synthesized using a consecutive electrodeposition−annealing process on a Ti foil substrate to obtain a self-supporting electrode (Methods; Figure 2a). Comparison samples, including RuO_2, W-RuO_2, and Ce-RuO_2 were prepared under identical conditions, differing only in the ionic types in the electrodeposition solution. Scanning electron microscopy (SEM) images of the pure RuO_2 sample (Figure S1) revealed a dense layer of uniformly sized RuO_2 spheres (100−200 nm). For the W-RuO_2 (Figure S2) and Ce-RuO_2 samples (Figure S3), SEM and energy-dispersive X-ray spectroscopy (EDS) results showed distinct RuO_2 spheres with uniform W and Ce doping, respectively. Morphological analyses of the (Ce W)-RuO_2 sample (Figure S4) confirmed uniform Ce distribution throughout the catalyst layer.

Further structural characterization using high-angle annular dark-field scanning transmission electron microscopy (HAADF-STEM) verified the uniform doping of W and Ce in the RuO_2 spheres of the (Ce W)-RuO_2 catalyst. The EDS area scans (Figure 2b) revealed homogeneous distributions of Ce and W within the RuO_2 spheres, with an approximate Ru:W atomic ratio of 3:1 and a Ce content of 2 at% (Figure S5). The HAADF-STEM line scans, SAED patterns, and HRTEM images (Figure S6) indicated that W and Ce were incorporated into the RuO_2 lattice in a homogeneous doping manner, rather than existing as separate oxide phases. Uniform doping was further corroborated by a slight increase in the lattice spacing of the RuO_2 (110) facet (Figure 2c), compared to the standard value of 0.318 nm (PDF#01−088−0286). This lattice expansion, attributed to the larger atomic radii of W (135 pm) and Ce (185 pm) compared to Ru (130 pm), confirms the successful doping, which is beneficial for the regulation of lattice oxygen activity and enhancement of LORM-OER stability.

To investigate the role of Ce doping in reactivating lattice oxygen activity, X-ray photoelectron spectroscopy (XPS) (Figure 2d) and Raman spectroscopy (Figure 2e) tests were performed. The decrease in Ru binding energy and Ru−O covalency after W doping was induced by the weakened orbital overlap between Ru and O due to lattice expansion. Subsequent doping of Ce restored both the Ru−O covalency and the binding energy. This can be attributed to the redox flexibility of Ce, which can reversibly transition between Ce^{4+} and Ce^{3+} states. Ce^{4+} functions as an electron acceptor, withdrawing excess electron density from adjacent Ru atoms, thereby increasing the Ru binding energy and strengthening Ru−O covalency. These effects were also supported by the Raman spectra. The clear E_g and A_{1g} peaks associated with Ru−O bonding confirmed the crystalline structure of RuO_2,[22,36] aligning with the TEM observations. In the W-RuO_2 catalyst, the Ru−O vibration peaks red-shifted relative to pure RuO_2, indicating elongated Ru−O bonding lengths and decreased Ru−O covalency, promoting AEM as the dominant reaction pathway.[11] The difference in redshifts between the two Raman shifts is caused by the different vibrational modes. The A_{1g} mode represents the stretching vibration of the Ru−O

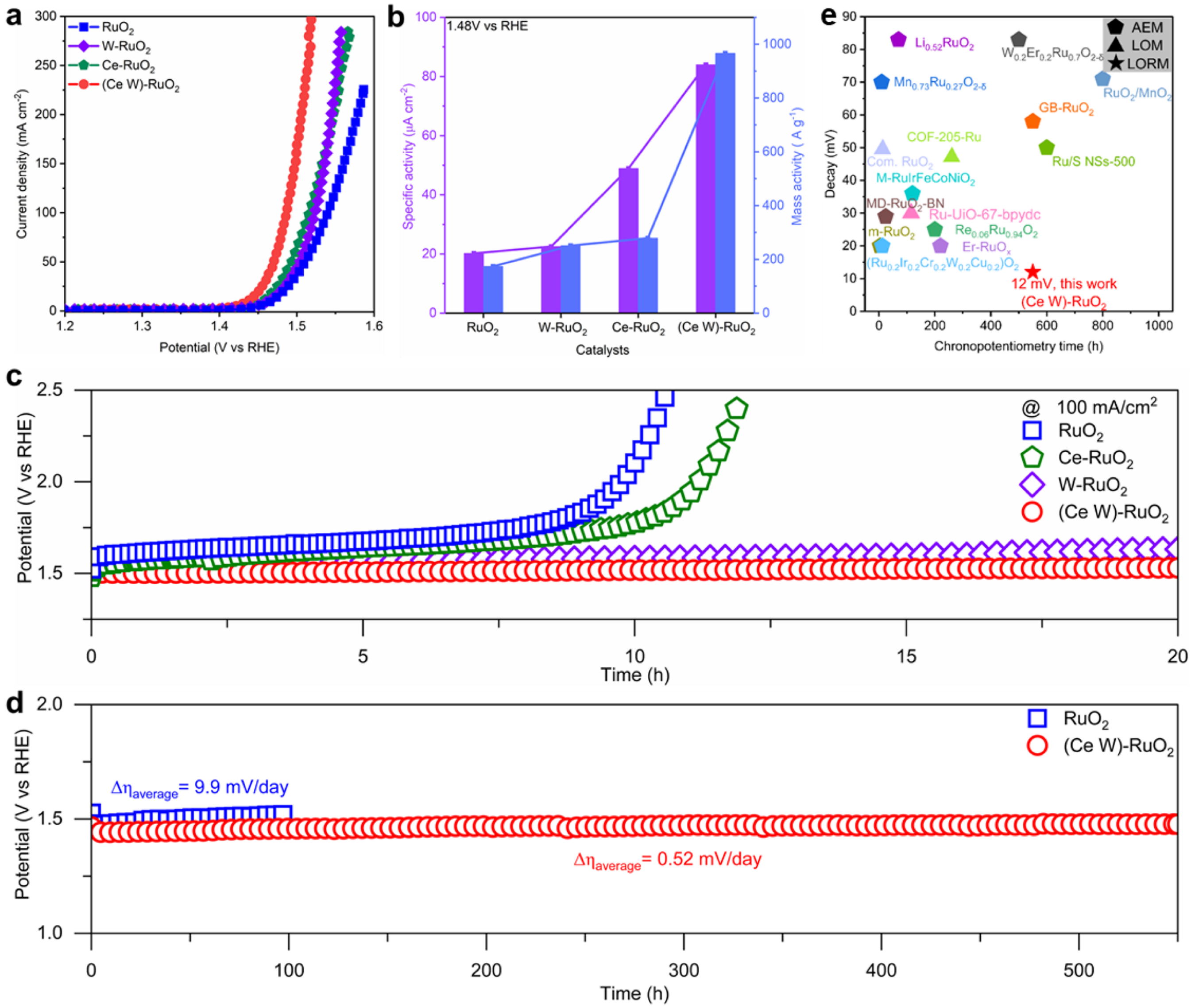

Figure 3. The evaluation of OER activity and stability of catalysts in 0.5 M H_2SO_4. (a) The apparent OER activity of electrodes with *iR* compensation. (b) The specific activity (left *y*-axis) and Ru mass activity (right *y*-axis) comparison of different electrodes. (c) The stability comparison of different electrodes by CP tests at the current density of 100 mA cm^{-2} with *iR* compensation. (d) The long-term CP test of RuO_2 and the (Ce W)-RuO_2 catalyst at a current density of 10 mA cm^{-2} with *iR* compensation. (e) The summarized comparison of acidic OER stability (details in Table S3).

bonds, while the E_g mode corresponds to bending vibrations involving the O−Ru−O angles. Upon Ce doping, this redshift disappeared, and the Ru−O vibration peaks returned to their original positions, suggesting that the (Ce W)-RuO_2 catalyst could retain the LOM pathway for OER. The OER mechanisms of the catalysts, determined through subsequent analyses, confirm that the (Ce W)-RuO_2 catalyst achieves stable LOM-OER performance due to its ability to reactivate and balance lattice oxygen activity effectively.

3.2. OER Performance Analysis. The linear sweep voltammetry (LSV) results (Figure 3a) demonstrated that the (Ce W)-RuO_2 electrode exhibited superior OER activity compared to RuO_2, Ce-RuO_2, and W-RuO_2 electrodes, achieving overpotentials of 206 mV at 10 mA cm^{-2} and 257 mV at 100 mA cm^{-2}. The reduced charge transfer resistance and the lower Tafel slope (48 mV dec^{-1}) facilitated faster reaction kinetics,[37,38] confirming the enhanced OER activity of the (Ce W)-RuO_2 electrode (Figure S7). To investigate the origin of the superior OER activity, the specific activity and Ru mass activity were evaluated. CV tests in a non-Faradaic potential window (Figure S8) showed similar double-layer capacitances (C_{DL}) and electrochemical active surface areas (ECSAs) across all electrodes, indicating that the superior OER activity of the (Ce W)-RuO_2 electrode was due to the higher specific activity rather than a larger ECSA (Figure S9). At 1.48 V vs RHE, the specific activity of the (Ce W)-RuO_2 electrode reached 86 μA cm^{-2} (left *y*-axis in Figure 3b), which is 1.3, 1.7, and 3.9 times that of the Ce-RuO_2, RuO_2 and W-RuO_2 electrodes, respectively. Using Ru mass loading determined by ICP-OES (Table S1), Ru mass activity was calculated (Figure S10). At 1.48 V vs RHE, the mass activity of the (Ce W)-RuO_2 electrode reached 966 A $g^{-1}{}_{Ru}$ (right *y*-axis in Figure 3b), which is 1.6, 2.1, and 4.0 times greater than the Ce-RuO_2, RuO_2, and W-RuO_2 electrodes, respectively. A comparison (Figure S11 and Table S2) highlighted the outstanding Ru mass activity of the (Ce W)-RuO_2 catalyst

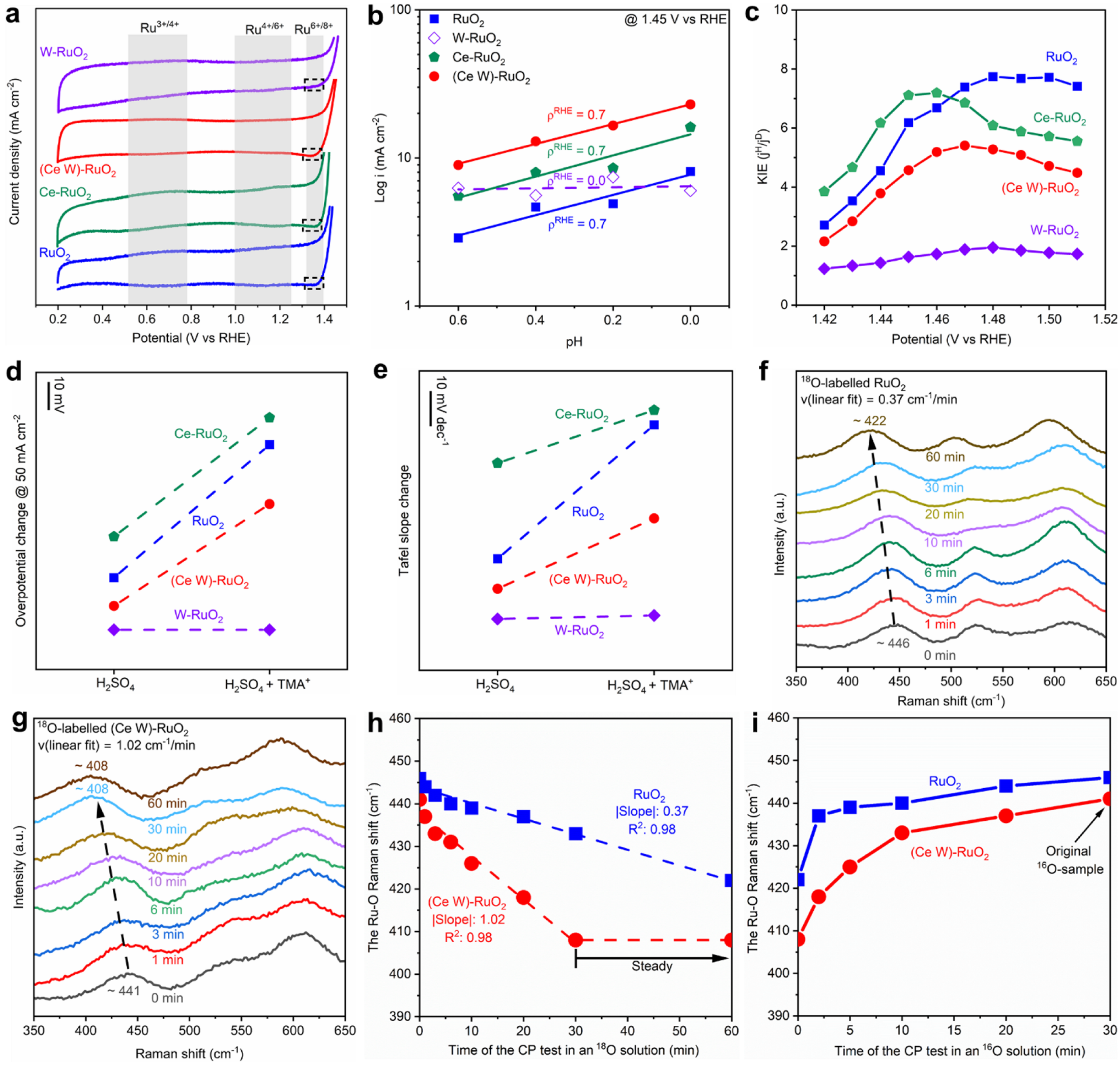

Figure 4. The OER mechanism analysis of catalysts. (a) The redox characteristics of Ru species analyzed by CV scans at a rate of 10 mV s^{-1}. (b) The proton reaction order. (c) The kinetic isotope effect. j^H and j^D refer to the current density in H_2O and D_2O solutions. (d) The overpotential increment at 50 mA cm^{-2} after adding TMA^+ to the electrolyte. (e) The Tafel slope increment after adding TMA^+ to the electrolyte. The variation of Raman spectra with ^{18}O refilling in the (f) RuO_2 catalyst and (g) (Ce W)-RuO_2 catalyst. (h) The linear fitting of the varied E_g Raman peak with the lattice oxygen refilling by ^{18}O. (i) The recovery of the E_g Raman peak with the release of lattice-^{18}O.

among recently reported catalysts, attributed to the lattice oxygen pathway, which bypasses the linear adsorption energy limitation of the AEM pathway.[15−18]

Stability, a critical criterion of LOM-based OER catalysts, was also evaluated. Chronopotentiometry (CP) tests at 100 mA cm^{-2} (Figure 3c) revealed that W-RuO_2 and (Ce W)-RuO_2 electrodes maintained stable OER performance, unlike the RuO_2 and Ce-RuO_2 electrodes, which became inactive after only 10 h. The extended stability measurement (Figure S12) showed that the (Ce W)-RuO_2 electrode operated for at least 80 h. Notably, the stability improvement observed for Ce-RuO_2 relative to RuO_2 was negligible, confirming our design principle (Figure 1g) that achieving a significantly larger $\Delta_1 REFI_{LO}$ than $\Delta_1 CONS_{LO}$ is challenging. Importantly, the (Ce W)-RuO_2 exhibited remarkable long-term stability during a 550-h CP test at 10 mA cm^{-2}, with an average potential increase of only 0.52 mV/day, significantly lower than 9.9 mV/day of RuO_2 (Figure 3d). The time-dependent ICP-OES tests (Figure S13) were performed during the continuous CP tests. After 21 h of continuous operation, the total dissolved Ru accounted for approximately 6.7% of the initial Ru content in the RuO_2 catalyst. In contrast, the (Ce W)-RuO_2 catalyst exhibited remarkable stability, with no clear dissolution signs of either W or Ce dopants, indicating that the doping elements remained immobilized within the catalyst structure and did not undergo noticeable dissolution during the test. This confirmed

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the high chemical stability of the dopants under the applied electrochemical conditions. The maximum detected Ru concentration was only 0.03 ppm, corresponding to a dissolution percentage of merely 0.8%. This clearly demonstrated the superior electrochemical stability of the (Ce W)-RuO_2 catalyst compared to its undoped counterpart. The excellent stability of the (Ce W)-RuO_2 catalyst was also verified by the retained SEM morphology (Figure S14) and OER activity (Figure S15) of the 550-h spent catalyst. Moreover, the performance tests under the simulated industrial settings (Figure S16) suggested that the (Ce W)-RuO_2 still demonstrated superior activity and stability compared to RuO_2, further validating the crucial role of improving lattice oxygen refilling in stabilizing LOM-based catalysts without compromising intrinsic activity. Compared to Ru-based catalysts in acidic solution reported in the literature (Figure 3e and Table S3), the (Ce W)-RuO_2 catalyst displayed exceptional OER stability due to its enhanced lattice oxygen refilling ability. For instance, the Ru-UiO-67-bpydc catalyst, which also operates via the LOM pathway, showed limited stability improvements by anchoring Ru atoms through stronger N−Ru bonding in MOF materials to reduce Ru dissolution.[19] However, its lattice oxygen still suffered continuous consumption without enhanced refilling, limiting the stability improvement. These results suggest that reinforcing lattice oxygen refilling is the most effective strategy for improving the stability of LOM-OER catalysts, as demonstrated in this work with the (Ce W)-RuO_2 catalyst.

3.3. OER Mechanism Analysis. To elucidate the reaction mechanism underlying the enhanced OER activity and stability of the (Ce W)-RuO_2 catalyst, electrochemical techniques and Raman spectroscopy were employed. For RuO_2 and Ce-RuO_2 samples (Figure 4a), redox peaks observed at 0.6, 1.1, and 1.35 V were attributed to Ru^{3+}/Ru^{4+}, Ru^{4+}/Ru^{6+}, and Ru^{6+}/Ru^{8+} transitions, respectively.[23,26] The Ru^{6+}/Ru^{8+} redox process, indicative of excessive oxidation leading to the formation of soluble *V_O-RuO_4^{2-} species, was suppressed in W-RuO_2 due to W doping, which improved OER stability by inhibiting lattice oxygen from directly forming O_2. In contrast, the Ru^{6+}/Ru^{8+} redox reappeared in the (Ce W)-RuO_2 catalyst, suggesting that its reaction mechanism aligns with that of RuO_2 and Ce-RuO_2.

pH dependence experiments provided further insights into the reaction pathway (Figure S17). By plotting the current density (j) at 1.45 V vs RHE on a logarithmic scale as a function of pH (Figure 4b), the proton reaction order ($\rho^{RHE} = \partial\log(j)/\partial pH$) for the W-$RuO_2$ sample was calculated to be 0, indicating no dependence on proton activity. In contrast, the ρ^{RHE} values for RuO_2, Ce-RuO_2, and (Ce W)-RuO_2 were all approximately 0.7, signifying a stronger dependence of OER activity on pH.[11,39] Similar trends were observed at 1.48 V vs RHE for larger current densities (Figure S18). To further evaluate the influence of proton activity, LSV tests were performed in H_2SO_4 (pH = 0) solution with H_2O and D_2O as solvents to assess the kinetic isotope effect (KIE) (Figure S19). As shown in Figure 4c, the KIE values for RuO_2, Ce-RuO_2, and (Ce W)-RuO_2 were significantly higher than that for W-RuO_2, indicating greater isotopic effects and more pronounced activity degradation when H_2O was replaced with D_2O. These results suggest that W-RuO_2 performs the OER predominantly through the AEM pathway, which involves a concerted proton−electron transfer (CPET) process. In contrast, RuO_2, Ce-RuO_2, and (Ce W)-RuO_2 follow the LOM pathway, which involves a high degree of nonconcerted proton−electron transfer (nCPET) process.[11,39] The nCPET process is responsible for the larger ρ^{RHE} and KIE values observed in these samples.

To further confirm that the (Ce W)-RuO_2 performed the OER via the LOM pathway, *O_2^{2-}-Ru species were captured using tetramethylammonium cation (TMA^+),[37,38,40] as detailed in Figure S20, with the corresponding LSV and Tafel results shown in Figures S21 and S22, respectively. The impact of TMA^+ on OER activity was evaluated by plotting the overpotential increment at the current densities of 50 mA cm^{-2} (Figure 4d) and 100 mA cm^{-2} (Figure S23), as well as the Tafel slope increment (Figure 4e). Significant increases in both overpotentials and Tafel slopes were observed for RuO_2, Ce-RuO_2, and (Ce W)-RuO_2 electrodes upon adding TMA^+, indicating the disruption of the LOM pathway. In contrast, the W-RuO_2 electrode showed negligible changes, consistent with its reliance on the AEM pathway. These results confirmed that the W-RuO_2 electrode performed the OER via the AEM pathway, while RuO_2, Ce-RuO_2, and (Ce W)-RuO_2 electrodes followed the LOM pathway.

To understand why the (Ce W)-RuO_2 electrode exhibits significantly improved OER stability compared to the RuO_2 electrode in acidic solution, despite both following the LOM pathway, ^{18}O isotope-labeling experiments and Raman spectroscopy were conducted. The synthesized ^{16}O electrodes (RuO_2 and (Ce W)-RuO_2) were first labeled with ^{18}O isotopes via consecutive CP tests for 60 min, and the Raman spectra were collected at the end of each CP test (Figure 4f,g). Subsequently, the ^{18}O-labeled electrodes underwent additional CP tests for 30 min in pure ^{16}O solution, and the Raman spectra were recorded after each CP test (Figure S24). Before each Raman measurement, the electrodes were rinsed thoroughly with deionized water (^{16}O) to remove the superficially adsorbed electrolyte. As shown in Figure 4f,g, the Raman peaks of both RuO_2 and (Ce W)-RuO_2 red-shifted with increasing reaction time due to the heavier mass of ^{18}O affecting the Ru−O vibration mode.[39,41,42] Notably, the redshift reached its maximum (33 cm^{-1}) after 30 min in the (Ce W)-RuO_2 electrode (Figure 4g) but continued to increase in the RuO_2 electrode up to 60 min, with a maximum of 24 cm^{-1} (Figure 4f). Since the redshift reflects O^{18} refilling from the electrolyte into the oxygen vacancies within the RuO_2 lattice, the observed redshift rate can be used to evaluate the rate of lattice oxygen refilling. Therefore, the Ru−O Raman shift over time was plotted, and the lattice oxygen refilling rate was estimated from the slope of a linear fit (Figure 4h), revealing a significantly larger slope for the (Ce W)-RuO_2 catalyst (1.02 cm^{-1}/min) compared to the RuO_2 catalyst (0.37 cm^{-1}/min). The much faster redshift for the (Ce W)-RuO_2 suggested an improved capability of lattice oxygen refilling, which effectively inhibits the formation of soluble *V_O-RuO_4^{2-} species, thereby improving the OER stability. In the subsequent ^{18}O-release CP tests (Figures S24 and 4i), Raman peaks gradually blue-shifted as ^{18}O was replaced by ^{16}O from the electrolyte. After 30 min, the Raman peaks of both electrodes returned to the positions of their original ^{16}O samples (dotted line in Figure S24), confirming the reversible lattice oxygen exchange. The combined results demonstrate that both RuO_2 and (Ce W)-RuO_2 perform the OER via the LOM pathway in acidic solutions. However, the improved lattice oxygen refilling capability of the (Ce W)-RuO_2 electrode enables enhanced stability. Our strategy directly

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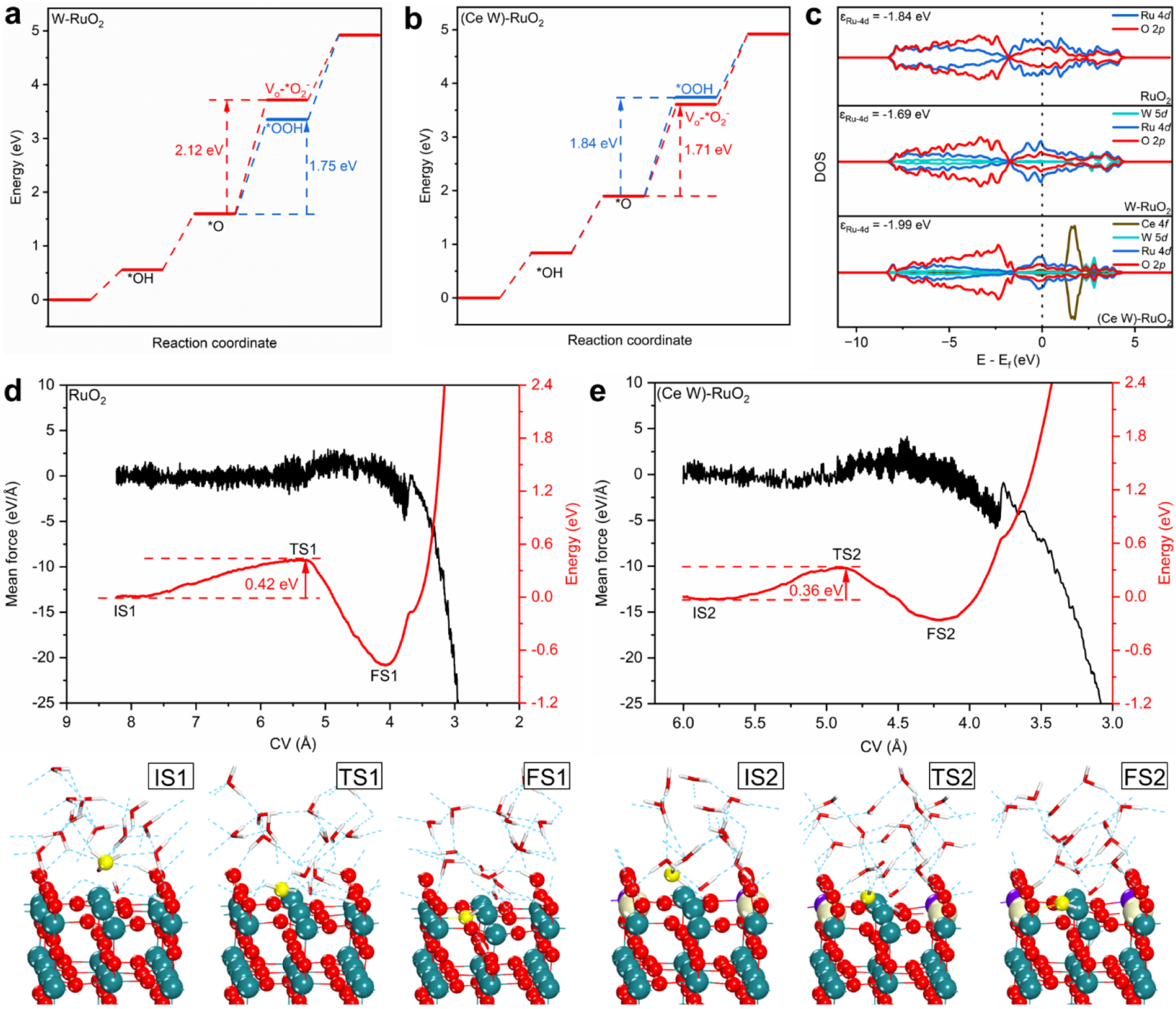

Figure 5. DFT calculations. The Gibbs free energy diagrams of the AEM and LOM pathways on the (a) W-RuO_2 catalyst and (b) (Ce W)-RuO_2 catalyst. (c) The density of states. Slow-growth simulation of the diffusion process of an H_2O molecule from the bulk solution to an oxygen-vacancy site on (d) the RuO_2 catalyst and (e) the (Ce W)-RuO_2 catalyst. The collective variables (CV) were set to the distance between O and (Ru Ru) or (Ru Ce) (blue ball: Ru atom, white ball: Ce atom, purple ball: W atom, red ball: O atom, red-and-white stick: H_2O molecule, yellow ball: a traced H_2O molecule).

improves LOM stability by promoting lattice oxygen refilling rather than replacing LOM with AEM, achieving stable acidic OER in RuO_2 via the proposed LORM and breaking the long-standing performance trade-off.

3.4. DFT Calculations. Density functional theory (DFT) calculations based on the rutile RuO_2 (110) surface were performed to elucidate the superior OER activity and stability of the (Ce W)-RuO_2 catalyst, attributed to the proposed LORM (Figures S25−S33). Because RuO_2 (110) with a different number of Ru vacancies exhibits different favorable OER routes,[43] the free energy diagrams of both AEM and LOM pathways were analyzed on RuO_2 (110) with one and two Ru vacancies (Figure S34). The energy barrier of the potential-determining step (PDS) in the LOM pathway decreased significantly compared to that in the AEM pathway as the number of Ru vacancies increased, rendering the LOM pathway more favorable for defective RuO_2. These results align with previous reports, confirming that practical RuO_2-based catalysts with defects typically perform OER via the LOM pathway.[19,43−45] For W-RuO_2 and (Ce W)-RuO_2, the PDSs correspond to the formation of *OOH intermediates (AEM, Figure 5a) and V_o-*O_2^- intermediates (LOM, Figure 5b), respectively, consistent with the experimental findings. The density of states (DOS) analysis (Figure 5c) revealed that the Ru *d*-band center for W-RuO_2 (−1.69 eV) shifted closer to the Fermi level compared to that of RuO_2 (−1.84 eV), thus facilitating the adsorption of *OOH species and promoting the AEM pathway. In contrast, the Ru *d*-band center for the (Ce W)-RuO_2 catalyst shifted further away from the Fermi level (−1.99 eV), reducing the adsorption of *OOH species and favoring the LOM pathway. Moreover, the calculated formation energy of oxygen vacancies (Figure S35) showed values of 2.31, 2.39, and 1.74 eV for RuO_2, W-RuO_2, and (Ce W)-RuO_2, respectively. These results indicated that (Ce W)-RuO_2 exhibited significantly enhanced ability to generate oxygen vacancies, thereby favoring the LOM pathway.

To investigate lattice oxygen refilling, a slow-growth simulation was performed for the H_2O molecule migrating from the bulk solution to an oxygen-vacancy site on RuO_2 and (Ce W)-RuO_2 catalysts (Figure 5d,e). It is worth noting that

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the doped Ce atom on the (Ce W)-RuO_2 catalyst exhibited a strong affinity for water molecules during the pre-equilibrium process, with a Ce−O(H_2O) distance of 1.65 Å shorter than that of the RuO_2 catalyst (Figure S36). During the slow-growth simulation, an H_2O molecule in bulk solution diffused and refilled into the oxygen vacancy site between Ce (or Ru) and Ru and then decomposed into *OH and *H intermediates, with the detailed migration process shown in Figure S37. The energy barrier of this process was determined to be 0.36 eV for the (Ce W)-RuO_2 catalyst (Figure 5e), significantly lower than 0.42 eV for the RuO_2 catalyst (Figure 5d), indicating faster and more efficient lattice oxygen refilling on the (Ce W)-RuO_2 catalyst. Together with the Raman spectroscopy results, the DFT analysis confirms that the enhanced lattice oxygen refilling capacity of the (Ce W)-RuO_2 catalyst is key to its superior OER stability in acidic environments while maintaining high intrinsic activity.

4. CONCLUSIONS

The typical RuO_2 catalyst operates the OER via the LOM, delivering high intrinsic activity but suffering from rapid deactivation due to fast lattice oxygen consumption and slow lattice oxygen refilling. Conventional stabilization strategies that shift the reaction mechanism from LOM to AEM inevitably sacrifice intrinsic activity, leading to a persistent trade-off between activity and stability. To overcome this challenge, we proposed the lattice oxygen refilling mechanism (LORM) in the (Ce W)-RuO_2 catalyst codoped with Ce and W. This approach achieves a balanced lattice oxygen consumption and refilling, effectively breaking the performance trade-off. This improved stability and activity are attributed to the oxygen buffering capacity of CeO_x, enabled by the Ce^{3+}/Ce^{4+} redox transition, which mitigates lattice oxygen depletion.

A key challenge in leveraging the oxygen buffering capacity of CeO_x lies in its poor selectivity toward lattice oxygen refilling. The redox properties of CeO_x facilitate both lattice oxygen consumption and refilling, making it difficult to achieve a steady balance. To address this, a W-doped RuO_2 catalyst was deployed as an intermediary to overcome the poor selectivity of CeO_x. The resulting (Ce W)-RuO_2 catalyst demonstrated a significantly enhanced lattice oxygen refilling capacity and achieved a stable lattice oxygen balance, as confirmed by Raman spectroscopy and slow-growth simulations. Consequently, the (Ce W)-RuO_2 catalyst exhibits exceptional stability under acidic conditions while retaining high intrinsic activity. These findings suggest that the (Ce W)-RuO_2 catalyst effectively breaks the performance trade-off via the proposed LORM, offering a robust solution for achieving an OER in energy conversion technologies.

■ ASSOCIATED CONTENT

Supporting Information

The Supporting Information is available free of charge at https://pubs.acs.org/doi/10.1021/acscatal.5c01382.

> SEM images, TEM images, and HAADF-STEM images; electrochemical performance; Raman spectra; atomic models for DFT calculations; reaction free energy and formation energy of oxygen vacancies; tables for comparing the performance of catalysts (PDF)

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Notes

The authors declare no competing financial interest.

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